

De novo Assembly

LONDON
SCHOOL of
HYGIENE
& TROPICAL
MEDICINE



This practical has 3 core objectives, to:

- Introduce genome assembly.
- Highlight the applications of assembly.
- Introduce assembly algorithms and principles.

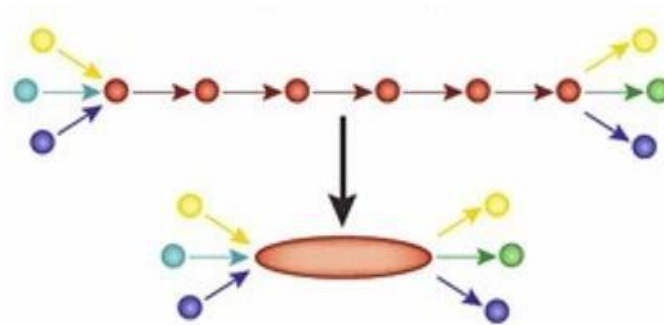
What is *De novo* Assembly?

***De novo* Assembly** is the process of reconstructing a complete genome from short fragments of DNA, without using a reference genome for guidance.

CTAGACCTACAGGATGCGCGACACGT
GGATGCGCGACACGTCGCATAT

(1)

Find overlaps
between fragments.



(2)

Assemble fragments into
contigs.



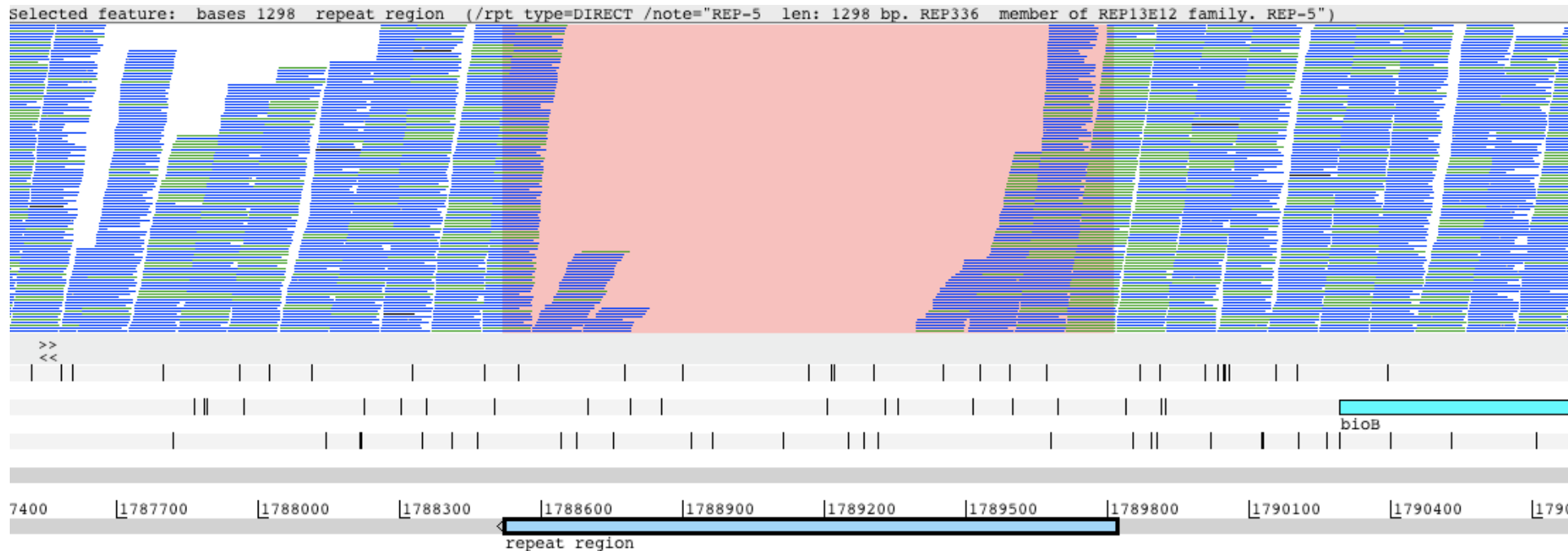
(3)

Assemble contigs into
scaffolds.

Why Assemble?

- No reference genomes for target organism.
- Highly variable / unstable regions in target organism.
- To investigate large structural variants, including:
 - Insertions
 - Deletions
 - Inversions
- To investigate novel transcripts in a transcriptome.

Why Assemble?

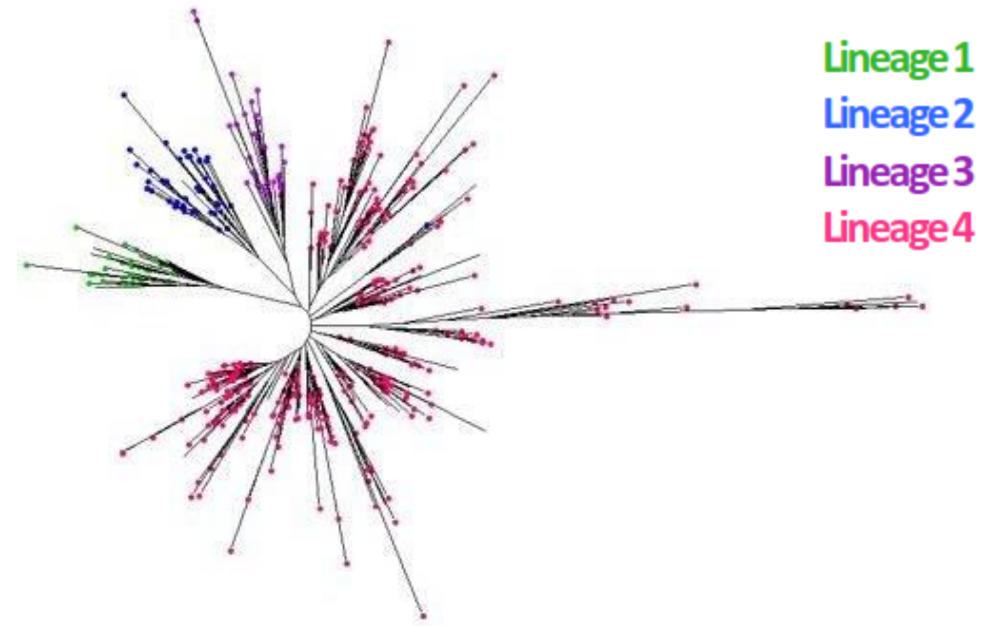


Overcome limitations of traditional mapping approaches which may fail to cover **repeat or highly variable regions**, specifically for short read data.

Assembly Applications

PE/PPE genes in *Mycobacterium tuberculosis* (TB) are typically excluded in downstream analysis due to:

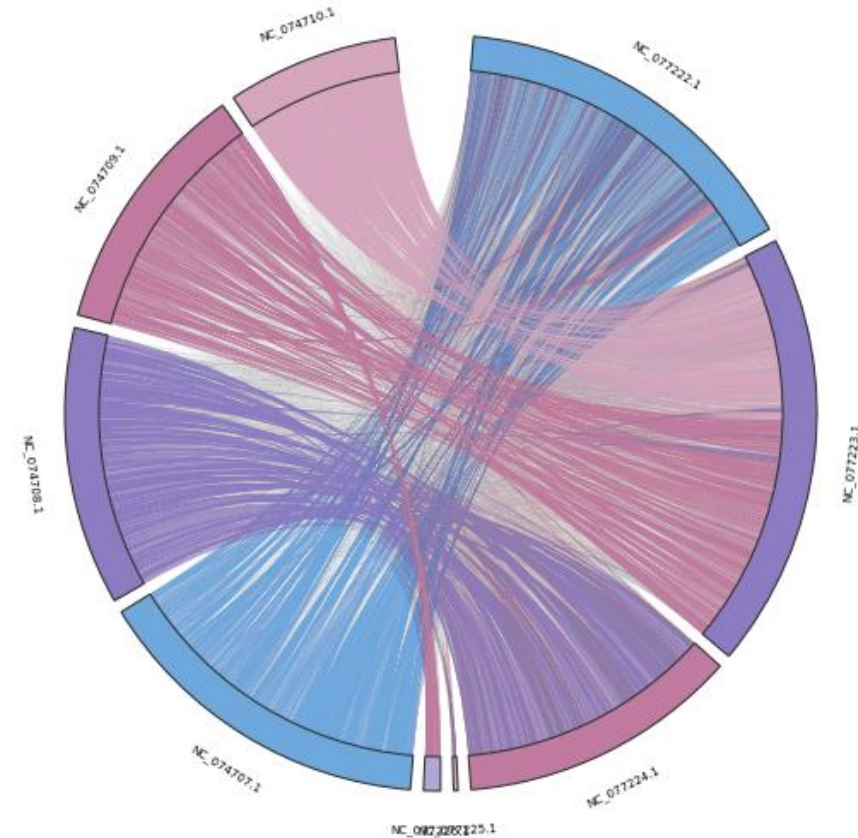
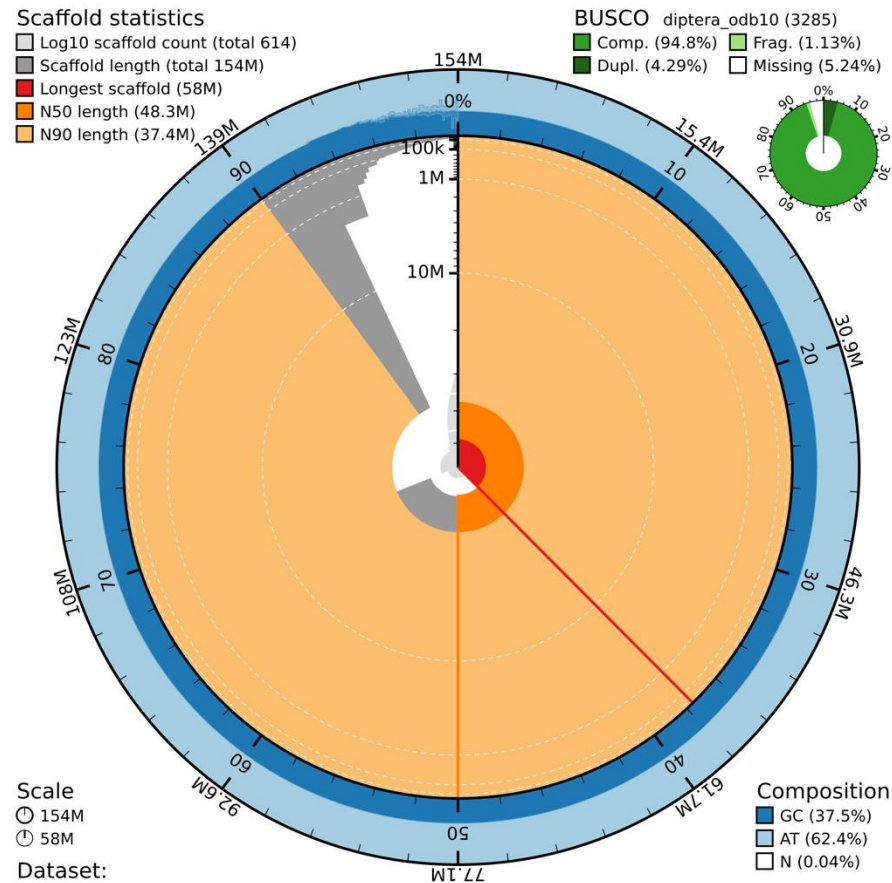
- High GC content
- Repetitive nature



An assembly approach allows us to reconstruct these gene enabling accurate downstream analysis across 500 clinical samples.

Assembly Applications

Our team is currently working on sandfly genome discovery



Genomic Jigsaw Puzzle



Complete genome = Assembled jigsaw puzzle.

Read = individual jigsaw piece.

De novo assembly can be thought of assembling the puzzle **without** knowing what the final picture should be.

Contrasts to **Mapping** where we know the final picture.

Assembly Challenges

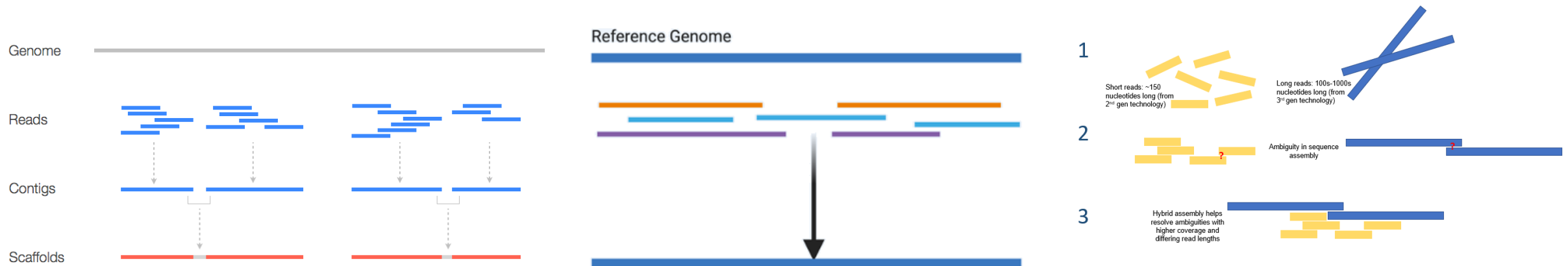
Challenges:

- Individual reads may fit together in more than one way and so need to optimise assembly to minimise introduction of errors.
- Require high levels of coverage for accurate and complete assembly.
- Requires stringent quality control to prevent error introduction.
- Computationally intensive.

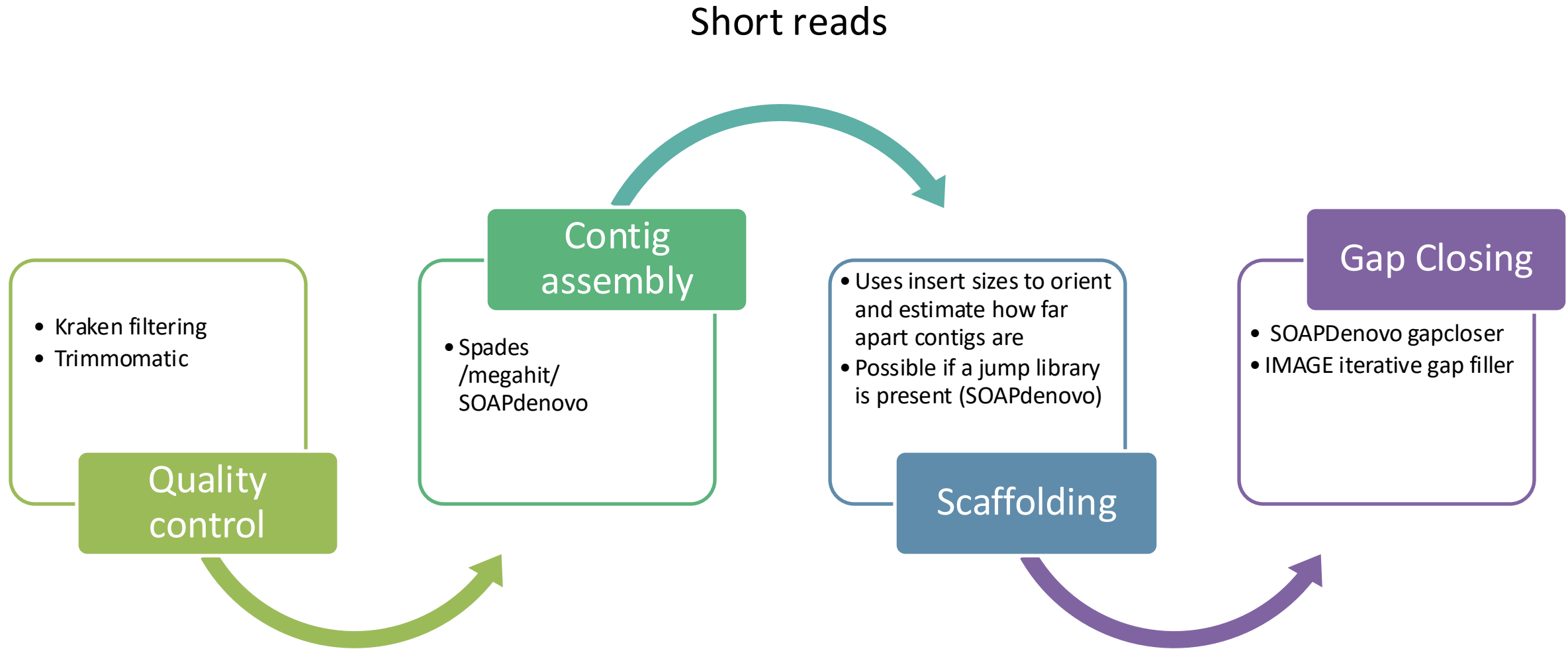
Assembly methods

There are three main ways to assemble a genome depending on the data available

1. Short read assembly
2. Long read assembly
3. Hybrid assembly



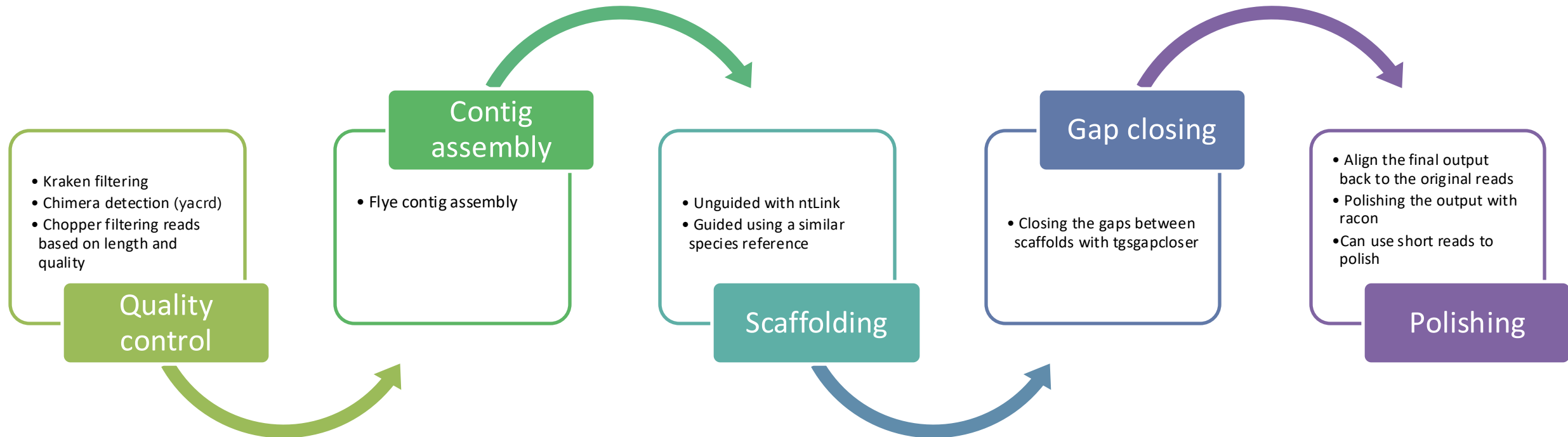
Assembly pipeline



Most tools are now outdated

Assembly pipeline

Long reads

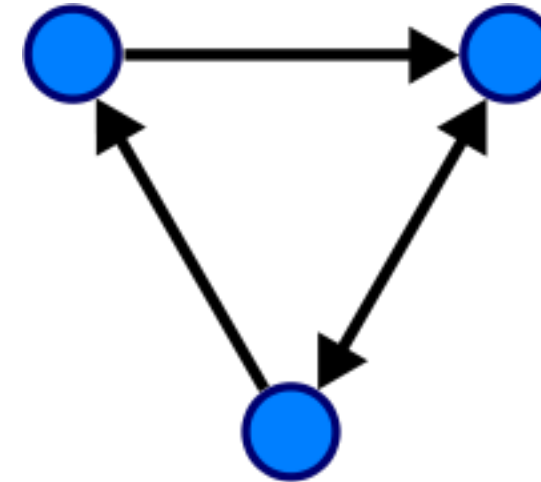


Can also use Hi-C data to assemble into chromosomes!

Assembly Algorithms

Each assembly tool uses a differing **Graph Theory** method.

A **Graph** is a mathematical structure that is used to represent objects and the connections between them.

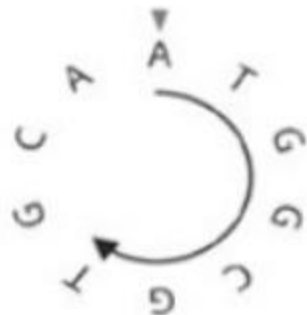


Node

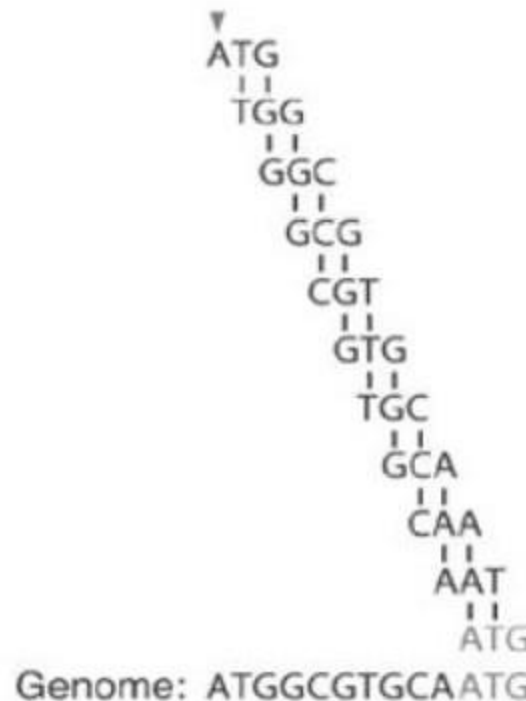


Edge

De Bruijn Graph Algorithm



(1)
Example DNA
Sequence
(e.g. read)



(2)
Split into kmer's via
sliding window

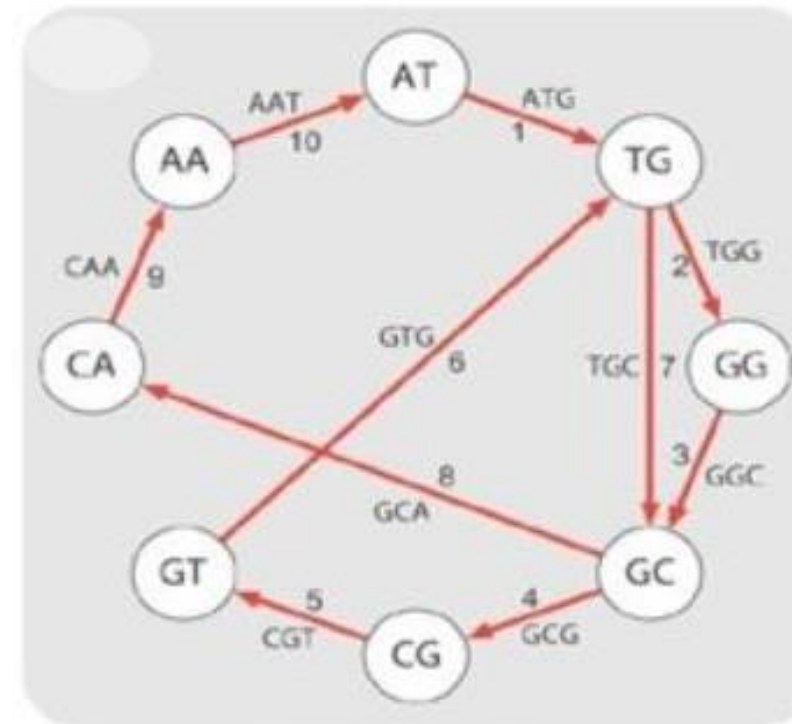


(3)
Split kmer into node
& edges

De Bruijn Graph Algorithm

(4) Constructed graph using all kmers.

(5) Find the path which uses each edge once, this will stitch k-mers together and create our **assembly**.



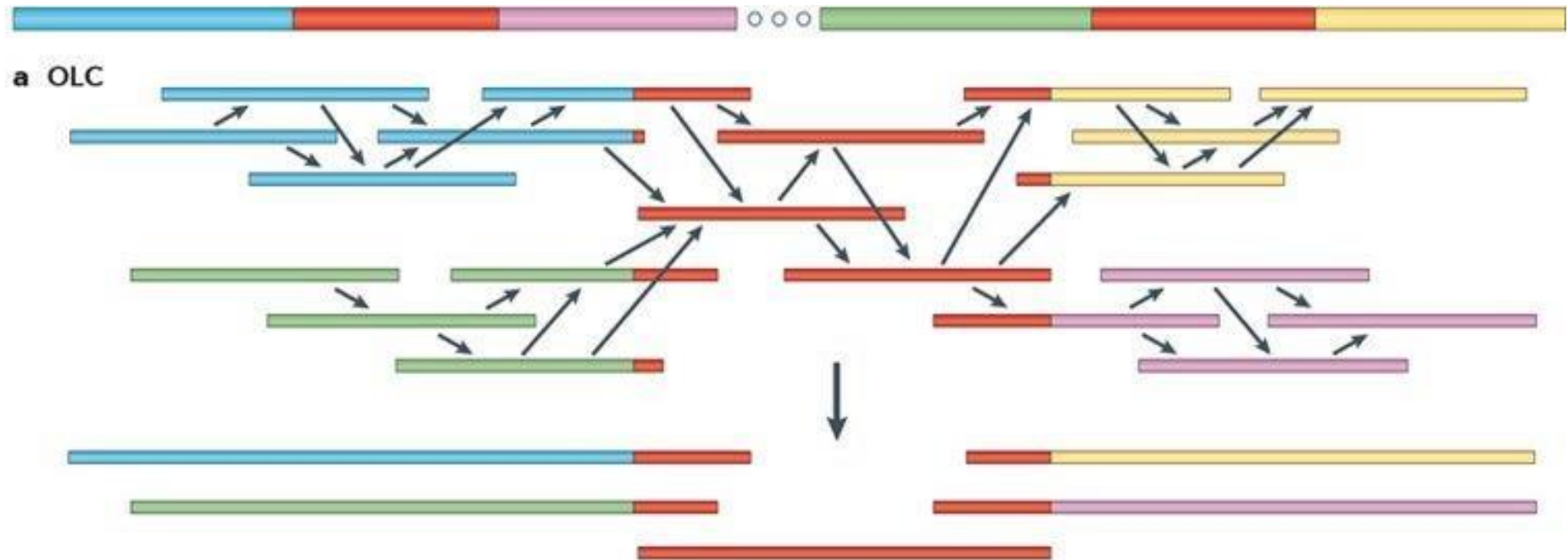
Eulerian cycle
Visit each edge once
(easier to solve)

OverlapLayout Consensus (OLC) Algorithm

Pairwise alignment
of all reads. Then a
graph is built using:

Reads = Nodes

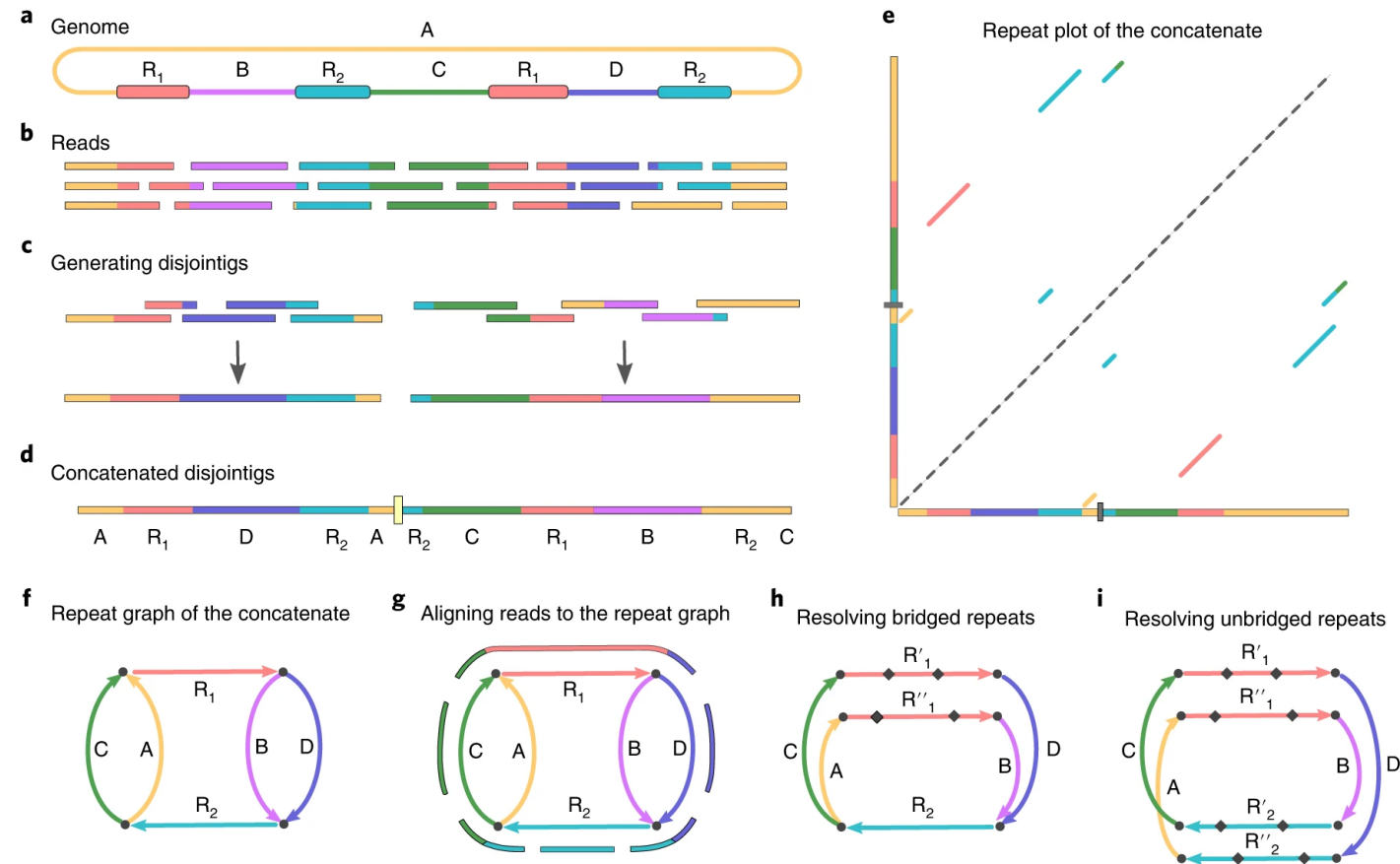
Overlap = Edges



From the construction of the graph we can begin to bundle **reads** into **contigs** as shown.

Repeat graphs

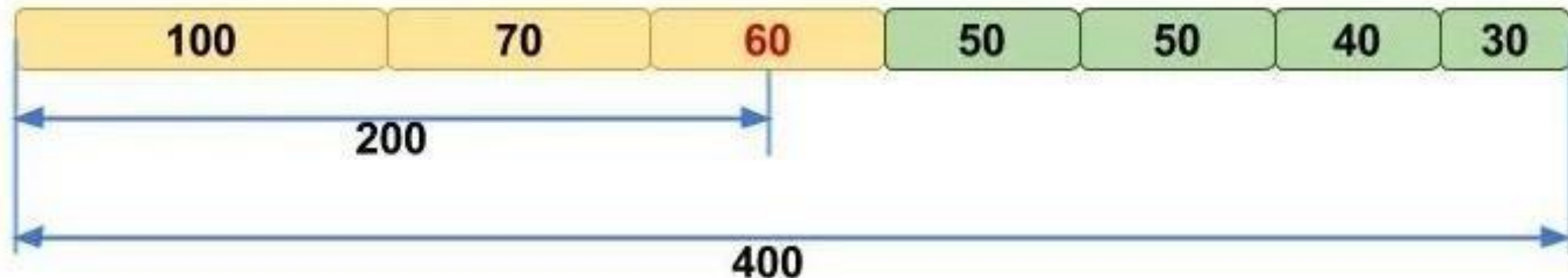
- Reads all data at once, creates disjointigs (accurate unambiguous segments of the genome)
- Arrange disjointigs in a linear sequence
- Creates a repeat plot to determine where the repeats are
- A repeat graph is created
- Reads are aligned to the repeat graph
- Repeats are resolved via bridged and unbridged repeats



Evaluating Assemblies

In the absence of a high-quality reference genome, assemblies are often evaluated on the basis of:

- **N50 Metric**= 50% of assembly is contained within contigs of this size or bigger



Evaluating Assemblies

In the absence of a high-quality reference genome, assemblies are often evaluated on the basis of:

- **BUSCO Score** = Identifies presence and completeness of single copy orthologs expected.
- The proportion of reads that can be assembled.

What about after assembly?

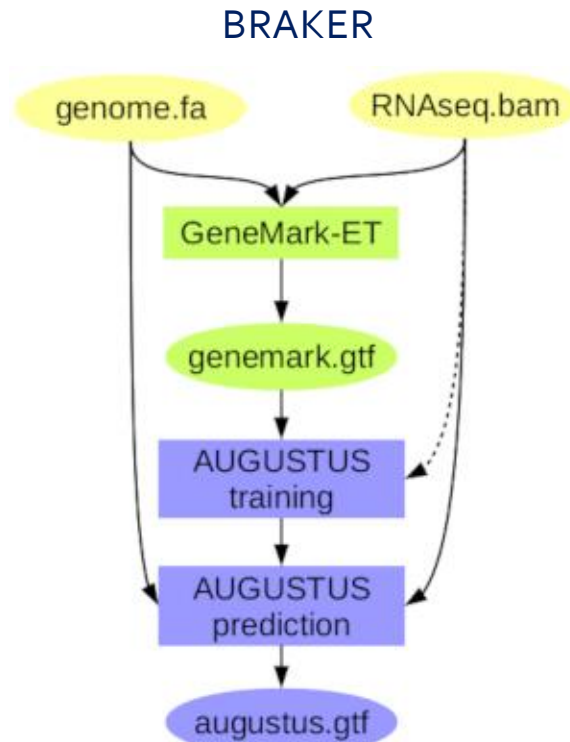


- Annotation
- Orthologs
- Mitochondrion discovery
- Comparative genomics

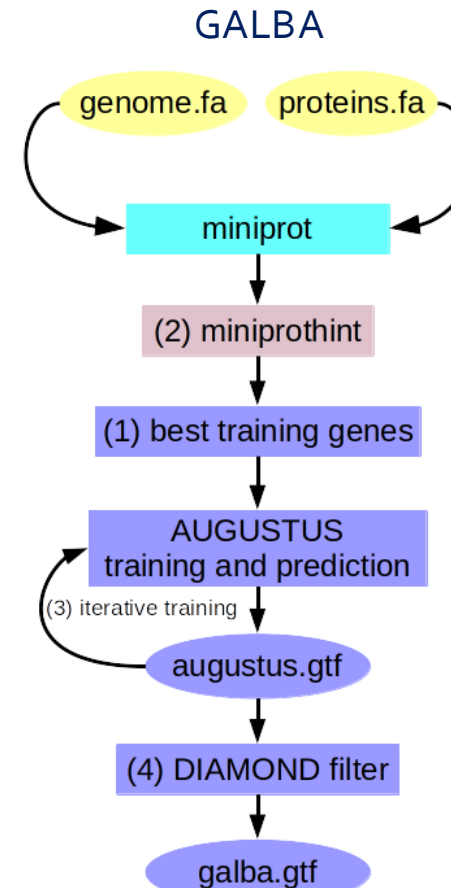
Reference Genome Annotation

Depends on the data available **

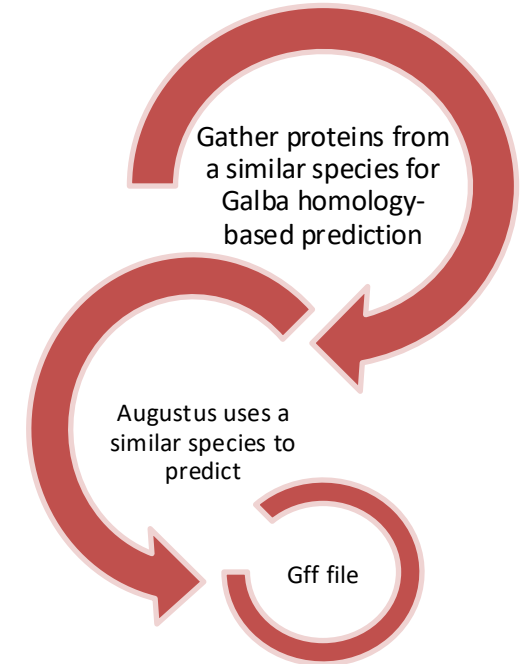
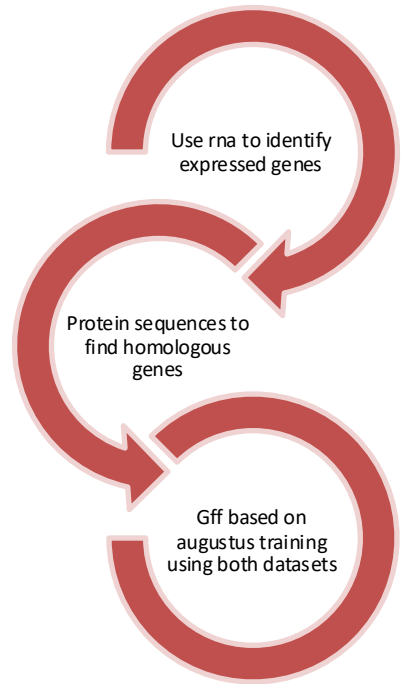
With RNA data



Without RNA data



Also homology approaches: liftoff, metaeuk, ncbi



Mitochondria discovery

*Requires a unique assembly approach

Reference mitochondria

Gather mitochondrial amino acid sequences from a similar species

Diamond makedb

Convert protein sequences into diamond database

Diamond blastx

Use diamond to align raw sequencing reads to the mitochondrial database

Filter

Filter hits by E-value threshold

Extract

Use these hits to extract mitochondrial reads from the original fastq

Re-assemble

Assemble the new mitochondrial genome with flye

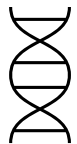
Comparative Genomics: Orthologs



Requires protein sequences from multiple similar species



Uses Markov clustering to identify groups of genes with common ancestry (orthologs)



Identifies core genes and gene families conserved across species



Helpful for annotating unknown genes by identifying orthologs with known functions

Practical Overview

Perform *de novo* assembly of *Mycobacterium tuberculosis* and look to identify and validate structural variants.

1. Short Read assembly
2. Long read assembly
3. Hybrid assembly